

Data Collection Techniques

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Data are facts, numbers, or observations gathered for a specific purpose.

Data Collection

Data collection is the process of gathering information from people, objects, or events.

Data Collection Techniques

Different **data collection techniques** are used depending on the type of information needed.

- ▶ Survey
- ▶ Observation
- ▶ Experiment
- ▶ Measurement
- ▶ Census and sample survey

Survey

A **survey** gathers information by asking questions through questionnaires or interviews.

Example: “How do Grade 7 students travel to school?”

Observation

An **observation** involves carefully watching and recording events or behaviors.

Example: A coach watches and records how many times players pass the ball in a game.

Experiment

An **experiment** is used to test a hypothesis by changing one factor and observing the results.

Example: A student changes the temperature of water to see how fast sugar dissolves.

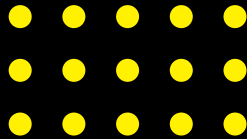
Measurement

A **measurement** collects numerical data such as length, mass, time, or temperature.

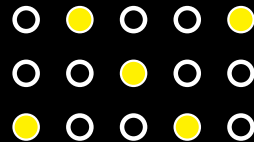
Example: A coach uses a stopwatch to time a runner in a 100-meter dash.

Census and Sample Survey

A **census** gathers information from every member of a population, while a **sample survey** collects data from only a part of the population.



Census



Sample survey

Choosing the appropriate data collection technique helps ensure that the collected data are accurate and useful.

Clue Words

If the situation...	Technique
asks people questions	Survey
watches and records events	Observation
changes one factor to test an effect	Experiment
finds length, mass, time, or temperature	Measurement
gathers data from every member	Census
gathers data from part of a group	Sample survey

Memory Tip

Ask, Watch, Test, Measure

Survey = **ask**

Observation = **watch**

Experiment = **test**

Measurement = **measure**

Census = **all** Sample survey = **some**

Examples

Example 1

Identify the most appropriate data collection technique.

“What kind of music do Grade 7 students like?”

Survey, because people are asked questions.

Example 2

Identify the most appropriate data collection technique.

“A student changes only the brand of paper towel to see which one absorbs more water.”

Experiment, because one factor is changed and the results are observed.

Example 3

Identify the most appropriate data collection technique.

“A teacher watches and records how students behave during group work.”

Observation, because behavior is watched and recorded.

Example 4

Identify the most appropriate data collection technique.

“A farmer determines the mass of harvested mangoes.”

Measurement, because mass is numerical data.

Example 5

Tell whether each situation uses a **census** or a **sample survey**.

(a) All 250 Grade 9 students answer a questionnaire. **Census**

(b) Only 30 randomly selected Grade 9 students answer a questionnaire. **Sample survey**

Example 6

Match the situation with the correct technique.

(A) Survey (B) Observation

(C) Experiment (D) Measurement

“A student records the room temperature every hour.”

(D) Measurement

Practice Exercises

Practice 1

Identify the most appropriate data collection technique.

“What is the favorite school subject of all Grade 7 students?”

Practice 2

Identify the most appropriate data collection technique.

“A scientist wants to know how the growth of a plant changes when different amounts of water are given.”

Practice 3

Identify the most appropriate data collection technique.

“A teacher records how many students arrive late every morning for two weeks.”

Practice 4

Identify the most appropriate data collection technique.

“A tailor determines the waist sizes of customers.”

Practice 5

Identify the most appropriate data collection technique.

“The school principal wants information from every student regarding internet access at home.”

Practice 6

Tell whether each situation uses a **census** or a **sample survey**.

A researcher asks all 520 Grade 7 students about their favorite sport.

Practice 7

Tell whether each situation uses a **census** or a **sample survey**.

A researcher interviews only 80 randomly selected Grade 7 students.

Practice 8

Match each situation with the correct technique.

- (A) Survey (B) Observation
(C) Experiment (D) Measurement

“A nurse records the height of 25 children.”

Practice 9

Which technique is best for determining the number of vehicles passing through a school gate between 7:00 a.m. and 8:00 a.m.?

Activities

Activity 1

Identify the most appropriate data collection technique.

“What is the favorite snack of all Grade 7 students?”

Activity 2

Identify the most appropriate data collection technique.

“A student investigates how sunlight affects the growth of bean plants.”

Activity 3

Identify the most appropriate data collection technique.

“A class officer records how many students borrow books from the library each day.”

Activity 4

Identify the most appropriate data collection technique.

“A carpenter determines the lengths of wooden boards.”

Activity 5

Identify the most appropriate data collection technique.

“A barangay captain gathers information from every household about waste disposal.”

Activity 6

Tell whether each situation uses a **census** or a **sample survey**.

A school asks every teacher to complete a satisfaction questionnaire.

Activity 7

Tell whether each situation uses a **census** or a **sample survey**.

A school interviews only 40 parents selected at random.

Activity 8

Match each situation with the correct technique.

- (A) Survey (B) Observation
(C) Experiment (D) Measurement

“A doctor records the body temperature of patients.”

Activity 9

Which technique is best for determining how many students enter the canteen during recess?

Thank you!